Chapter 9C

Altered Mental Status, Syncope, Seizures, and Acute Neurologic Triage

Medvale Internal Medicine - Neurology for Internal Medicine

This chapter is a content-heavy framework for the common internal medicine presentations where the first question is not the final neurologic diagnosis, but whether the patient is unstable, metabolically poisoned, infected, seizing, postictal, syncopal from cardiac disease, delirious, comatose, or carrying a structural brain emergency. It uses the uploaded neuro sections on altered mental status, coma, syncope, and seizures as coverage guides.

1. Big picture: the four emergency questions

Altered mental status, syncope, seizure, and coma overlap at the bedside because patients often present after an event rather than during it. The internist must first decide whether the episode was a transient loss of consciousness from cerebral hypoperfusion, an epileptic event from abnormal cortical discharge, delirium from systemic illness, intoxication or withdrawal, or a structural neurologic catastrophe such as hemorrhage, mass, hydrocephalus, or meningitis.

The first several minutes are about reversible danger: airway, breathing, circulation, glucose, oxygenation, temperature, trauma, medications, intoxication, withdrawal, infection, and raised intracranial pressure. A normal CT, a normal EEG, or a normal lab panel does not automatically close the case; the clinical time course and witness history remain central.

First branch point: what syndrome is this?

Presentation	Core definition	Key immediate concern
Delirium	Acute fluctuating disturbance in attention, awareness, and cognition, usually caused by a medical disturbance or drug effect.	Underlying illness, infection, medication toxicity, hypoxia, metabolic disorder, withdrawal, urinary retention, constipation, pain.
Coma	Depressed consciousness with inability to be aroused normally.	Airway protection, structural lesion, metabolic/toxic cause, CNS infection, seizure/nonconvulsive status, increased ICP.
Syncope	Transient loss of consciousness and postural tone due to reduced cerebral blood flow, followed by rapid recovery.	Cardiac arrhythmia, structural heart disease, PE, hemorrhage, exertional syncope, injury.
Seizure	Transient signs/symptoms from abnormal excessive or synchronous neuronal activity.	Status epilepticus, hypoglycemia, hyponatremia, CNS infection, hemorrhage, intoxication/withdrawal, pregnancy/eclampsia.

2. Altered mental status: MOVE STUPID plus bedside stabilization

The uploaded source uses the mnemonic MOVE STUPID for altered mental status: metabolic, oxygen, vascular, endocrine, seizure, trauma, uremia, psychiatric, infectious, and drugs. This is useful because AMS is not a diagnosis. It is a syndrome that often reflects a systemic process in the brain. The workup should begin with the most reversible and dangerous causes, then narrow based on tempo and associated findings.

MOVE STUPID expanded

Category	Examples	Clues
Metabolic	Hypercalcemia, hepatic encephalopathy, hyponatremia, hypernatremia, acid-base disorder, hypoglycemia, hyperglycemia.	Diffuse confusion, asterixis, dehydration, abnormal electrolytes, liver/renal disease.
Oxygen / ventilation	Hypoxemia, hypercapnia, CO poisoning, severe anemia, shock.	Low O2 saturation, COPD, sedatives, headache, cyanosis, respiratory distress.
Vascular	Stroke, ICH, SAH, subdural hematoma, hypertensive encephalopathy.	Focal deficit, sudden onset, headache, trauma, anticoagulation, severe BP.
Endocrine	Hypoglycemia, thyroid storm, myxedema coma, adrenal crisis.	Temperature extremes, glucose abnormality, bradycardia/tachycardia, hypotension.

Category	Examples	Clues
Seizure	Postictal state, nonconvulsive status epilepticus.	Witnessed convulsions, tongue bite, incontinence, fluctuating unexplained coma.
Trauma	Concussion, intracranial bleed, cervical spine injury.	Falls, bruising, anticoagulants, intoxication, unwitnessed event.
Uremia	Renal failure, dialysis disequilibrium, severe acidemia.	High BUN/Cr, pericarditis, platelet dysfunction, pruritus.
Psychiatric	Primary psychosis, severe depression, catatonia, conversion.	Diagnosis of exclusion when medical causes are reasonably addressed.
Infectious	Sepsis, meningitis, encephalitis, UTI in older adults, pneumonia.	Fever may be absent in older/immunocompromised patients; look for source.
Drugs	Opioids, sedatives, anticholinergics, alcohol, withdrawal, polypharmacy.	Pupils, respiratory pattern, medication list, toxidrome, response to antidote.

Initial AMS orders and actions

- Airway, oxygenation, ventilation, circulation, temperature, cardiac monitor, IV access.
- Point-of-care glucose immediately; give dextrose if hypoglycemic and do not wait for formal labs.
- Consider thiamine before glucose in malnourished or alcohol-use patients, but do not delay correction of life-threatening hypoglycemia.
- CBC, CMP, calcium, magnesium, phosphate when relevant, venous/arterial blood gas if ventilation or acid-base disorder is suspected.
- Urinalysis, cultures, chest imaging, lactate, lumbar puncture, toxicology, pregnancy test, thyroid/adrenal testing, or CT/MRI based on clinical context.
- Review medications for sedatives, anticholinergics, opioids, serotonergic agents, dopamine blockers, lithium, anticonvulsants, insulin/sulfonylureas, and recent medication changes.

> AMS trap: older adults may present with infection, urinary retention, fecal impaction, medication toxicity, pain, hypoxia, or dehydration as confusion without classic localizing symptoms. Hypoactive delirium is easy to miss because the patient appears quiet rather than agitated.

3. Delirium versus dementia versus primary psychiatric disease

Delirium is acute, fluctuating, and attention-based. Dementia is chronic and progressive with preserved alertness until late stages. Primary psychiatric disease may cause disorganized thought or hallucinations, but it usually does not cause a fluctuating level of consciousness or inattention in the same way delirium does. The source emphasizes that delirium is typically reversible if the underlying cause is found, while dementia is usually progressive.

Delirium vs dementia vs psychosis

Feature	Delirium	Dementia	Primary psychosis
Onset	Hours to days; abrupt.	Months to years; insidious.	Variable; often subacute/chronic.
Course	Fluctuating, often worse at night.	Slow progressive decline.	May fluctuate with stress but not classically attention-based.
Attention	Impaired; cannot sustain focus.	Often preserved until later.	Usually preserved unless severely disorganized.
Consciousness	Altered, fluctuating arousal.	Usually preserved.	Usually preserved.
Hallucinations	Often visual or mixed, frightening.	May occur in Lewy body disease.	Often auditory; delusions prominent.
Treatment target	Find medical cause and remove precipitants.	Assess reversible contributors and support function.	Psychiatric evaluation after medical causes excluded.

Delirium management principles

Management is primarily cause-directed and environmental. Avoid reflexively sedating delirious patients unless they pose an immediate danger to themselves or staff. Antipsychotics may be used cautiously for severe agitation, but they do not treat the underlying delirium and can worsen QT prolongation, parkinsonism, aspiration risk, or neuroleptic sensitivity in Lewy body disease.

• Treat the trigger: infection, hypoxia, retention, constipation, pain, withdrawal, medication toxicity, metabolic derangement.

• Reorient frequently; restore glasses/hearing aids; normalize sleep-wake cycle; mobilize early.

• Avoid benzodiazepines except for alcohol/benzodiazepine withdrawal or selected indications.

• Reduce deliriogenic medications: anticholinergics, sedative-hypnotics, opioids when possible, dopamine blockers when inappropriate.

• Use family/caregivers to establish baseline cognition and function.

4. Coma and depressed consciousness

Coma is depressed consciousness to the point that the patient is not normally arousable. The source separates structural causes from global metabolic/toxic causes. Structural lesions often produce focal findings, asymmetric pupils, asymmetric motor responses, or abnormal brainstem reflexes. Metabolic and toxic causes more often produce symmetric findings, though severe metabolic illness can still produce focal-appearing abnormalities.

Coma clues

Finding	Suggests	Interpretation
Unilateral dilated pupil	CN III compression or herniation.	Structural emergency until proven otherwise.
Pinpoint pupils	Opioids or pontine lesion.	Check respiratory rate; naloxone if opioid toxicity possible.
Reactive symmetric pupils	Metabolic/toxic cause more likely.	Does not exclude structural disease.
Decorticate posture	Lesion above red nucleus / severe bilateral hemisphere injury.	Serious brain injury.
Decerebrate posture	Brainstem involvement or severe diffuse injury.	Worse prognostic sign.
Absent oculocephalic response	Brainstem dysfunction.	Do not perform if cervical spine injury suspected.
Fever and neck stiffness	Meningitis, encephalitis, SAH.	Blood cultures/antimicrobials and LP or imaging pathway.
No motor response with preserved vertical eye movements	Locked-in syndrome.	Can mimic coma; patient may be aware.

Glasgow Coma Scale: use it serially

GCS components

Component	Score range	High-yield meaning
Eye opening	1-4	Spontaneous eye opening is 4; no eye opening is 1.
Verbal response	1-5	Oriented is 5; incomprehensible sounds or no verbal response indicate severe impairment.
Motor response	1-6	Obeys commands is 6; localizes pain is better than withdrawal; abnormal posturing is severe.

The absolute number is less useful than the trajectory. A falling GCS, new pupillary asymmetry, worsening headache, vomiting, or abnormal respirations should trigger urgent reassessment for intracranial catastrophe, hypoxia, hypercapnia, seizure, hypoglycemia, or shock.

5. Syncope: transient cerebral hypoperfusion

Syncope is transient loss of consciousness and postural tone caused by reduced cerebral blood flow with rapid spontaneous recovery. The source emphasizes that cardiac syncope has the worst prognosis and should be ruled out first. Seizure can cause loss of consciousness but is not technically syncope because the mechanism is abnormal neuronal discharge rather than cerebral hypoperfusion.

Syncope categories

Type	Typical story	Clinical danger
Vasovagal / reflex	Trigger such as pain, emotion, standing, heat; prodrome of nausea, warmth, diaphoresis, pallor, tunnel vision.	Usually benign but injury can occur.

Type	Typical story	Clinical danger
Orthostatic hypotension	Occurs after standing; associated with volume depletion, autonomic failure, diabetic neuropathy, Parkinson disease, medications.	Falls, dehydration, medication toxicity, occult bleeding.
Cardiac arrhythmia	Sudden syncope without prodrome; may occur supine or with exertion; palpitations may precede.	Sudden death risk; requires ECG/rhythm evaluation.
Structural cardiac	Exertional syncope, murmur, chest pain, dyspnea; aortic stenosis, HCM, pulmonary hypertension.	High-risk; urgent cardiac evaluation.
Pulmonary embolism	Syncope with dyspnea, pleuritic pain, tachycardia, hypoxemia, risk factors.	Life-threatening obstructive shock.
Cerebrovascular	Rare as isolated syncope; vertebrobasilar TIA can cause drop attacks with other posterior symptoms.	Look for diplopia, dysarthria, ataxia, focal deficits.

Syncope evaluation algorithm

- History before, during, and after the event: posture, prodrome, exertion, palpitations, chest pain, dyspnea, witness movements, duration, recovery.
- Medication review: antihypertensives, diuretics, vasodilators, QT-prolonging drugs, sedatives, alcohol.
- Orthostatic vitals, cardiac and neurologic exam, trauma exam.
- ECG for all patients with syncope.
- Pregnancy test when relevant; hemoglobin if bleeding is possible; troponin/PE workup only when history or exam suggests.
- Echocardiography if structural heart disease is suspected; ambulatory rhythm monitoring if intermittent arrhythmia is plausible.
- Do not order broad neurologic testing for simple vasovagal syncope with normal neurologic exam.

> High-risk syncope features: exertional syncope, syncope while supine, no prodrome, palpitations before event, family history of sudden death, abnormal ECG, structural heart disease, severe anemia/bleeding, hypotension, dyspnea, or chest pain.

Syncope versus seizure

Syncope vs seizure

Feature	Syncope	Seizure
Mechanism	Transient global cerebral hypoperfusion.	Abnormal cortical electrical discharge.
Prodrome	Lightheadedness, warmth, nausea, visual dimming, diaphoresis.	Aura may include smell, déjà vu, rising epigastric sensation, focal sensory symptoms.
Movements	Brief myoclonic jerks can occur after LOC.	Tonic-clonic activity often longer and rhythmic.
Duration of LOC	Usually brief.	Often 1-2 minutes for generalized tonic-clonic seizure.
Recovery	Rapid return to baseline unless injury or prolonged hypotension.	Postictal confusion, sleepiness, headache, myalgias.
Tongue bite	Rare; if present often tip.	Lateral tongue bite supports seizure.
Incontinence	Can occur.	Can occur; not specific alone.

6. Seizures and epilepsy

A seizure is a transient event caused by abnormal excessive or synchronous neuronal activity. Epilepsy is a tendency toward recurrent unprovoked seizures. A first seizure should prompt careful distinction between provoked seizure, unprovoked seizure, syncope with convulsive movements, psychogenic nonepileptic seizures, migraine, TIA, and movement disorder. The source organizes seizure causes around metabolic/electrolyte abnormalities, mass lesions, missing drugs, withdrawal, infection, ischemia, and increased intracranial pressure.

Common seizure precipitants

Category	Examples	Clue
Metabolic/electrolyte	Hyponatremia, hypoglycemia, hyperglycemia, hypocalcemia, uremia, hepatic failure.	Abnormal labs; diffuse encephalopathy.

Category	Examples	Clue
Structural	Stroke, tumor, hemorrhage, abscess, cortical scar, traumatic brain injury.	Focal deficit, headache, cancer, trauma.
Infection/inflammation	Meningitis, encephalitis, brain abscess, autoimmune encephalitis.	Fever, neck stiffness, CSF abnormalities, psychiatric/autonomic signs.
Toxic/withdrawal	Alcohol, benzodiazepine/barbiturate withdrawal, cocaine, lithium, theophylline, bupropion, tramadol.	Medication/substance history, autonomic activation.
Pregnancy	Eclampsia.	Hypertension, proteinuria, headache, visual symptoms, RUQ pain.
Medication nonadherence	Missed antiseizure medications.	Low drug level or history of missed doses.

Seizure classification made usable

Seizure types

Type	Clinical pattern	Common treatment anchor
Focal aware	Focal motor, sensory, autonomic, or psychic symptoms with preserved awareness.	Work up focal lesion; antiseizure medication if recurrent/high risk.
Focal impaired awareness	Altered awareness, automatisms, postictal confusion; often temporal lobe.	EEG/MRI; antiseizure therapy.
Focal to bilateral tonic-clonic	Begins focally then generalizes.	Treat as focal epilepsy with generalization.
Generalized tonic-clonic	Loss of consciousness, tonic phase, clonic jerking, postictal state.	Evaluate provoked causes; antiseizure therapy depending on recurrence risk.
Absence	Brief staring spells, eyelid fluttering, no postictal confusion; often childhood.	Ethosuximide classic for pure absence.
Myoclonic	Brief shock-like jerks.	Consider juvenile myoclonic epilepsy or metabolic/toxic causes.

First seizure workup

The workup is driven by whether the event was provoked and whether there are focal signs, persistent AMS, trauma, immunosuppression, fever, cancer, anticoagulation, pregnancy, or suspected CNS infection. AAN/AES guidance for an unprovoked first seizure emphasizes recurrence risk discussion and individualized treatment decisions. EEG supports diagnosis and classification but should not be used as a stand-alone exclusion test when the history is convincing.

- Glucose and electrolytes, especially sodium and calcium; renal and hepatic function when relevant.
- Pregnancy test when relevant.
- Toxicology and antiseizure medication levels when suggested by history.
- Noncontrast CT urgently if trauma, focal deficit, persistent AMS, anticoagulation, immunosuppression, cancer, fever, severe headache, or concern for hemorrhage/mass.
- MRI brain with epilepsy protocol when unprovoked seizure, focal features, abnormal neuro exam, or concern for structural epilepsy.
- EEG to support diagnosis, classify seizure type, and estimate recurrence risk.
- LP if meningitis, encephalitis, SAH, or inflammatory disease is suspected after considering imaging safety.

Status epilepticus: do not wait

Convulsive status epilepticus is a neurologic emergency. The practical threshold is a seizure lasting about 5 minutes or recurrent seizures without return to baseline. Treatment is staged: stabilize airway/oxygen/glucose, give benzodiazepine promptly, load a longer-acting antiseizure medication, and escalate to anesthetic therapy/intubation for refractory status.

Status epilepticus framework

Phase	Action	Pearl
0-5 minutes	ABCs, oxygen, IV access, glucose, thiamine if indicated, labs, protect from injury.	Do not put anything in the mouth.
Established convulsive seizure	Give full-dose benzodiazepine promptly.	Underdosing benzodiazepines is a common cause of persistent seizures.
Persistent seizure	Load antiseizure medication such as levetiracetam, fosphenytoin/phenytoin, or valproate depending on context.	Choice depends on comorbidities, pregnancy, liver disease, interactions.
Refractory	ICU, continuous EEG, anesthetic infusion, intubation/ventilation.	Consider nonconvulsive status after convulsions stop but coma persists.

Seizure trap: after convulsions stop, a persistently altered patient may still be in nonconvulsive status epilepticus. Consider EEG when mental status does not improve as expected.

7. Psychogenic nonepileptic seizures and mimics

Psychogenic nonepileptic seizures are real episodes of altered movement, sensation, or awareness that are not caused by epileptic cortical discharges. Diagnosis should be careful and respectful, usually requiring video EEG when uncertainty persists. Do not label an event psychogenic simply because it looks unusual. Syncope, frontal lobe seizures, movement disorders, panic, migraine, and metabolic events can all be misleading.

Mimic clues

Event	Clue	Next step
Convulsive syncope	Brief jerks after faint, rapid recovery, orthostatic/vasovagal trigger.	Syncope evaluation and ECG.
PNES	Prolonged fluctuating event, asynchronous movements, eye closure, emotional trigger; not diagnostic alone.	Neurology referral and video EEG when needed.
Migraine aura	Spreading positive visual/sensory symptoms; headache may follow.	Consider first/worst/vascular red flags.
TIA	Negative focal symptoms such as weakness or vision loss.	Vascular evaluation if focal ischemia possible.
Rigors	Shaking chills with preserved awareness.	Search for infection/sepsis.

8. Coma, brain death, and persistent vegetative state

The source distinguishes brain death from persistent vegetative state. Brain death requires irreversible absence of brain and brainstem function, coma, absent brainstem reflexes, and apnea despite adequate oxygenation and ventilation, with confounders such as hypothermia, drug intoxication, and severe metabolic derangement excluded. Persistent vegetative state has wakefulness without awareness; patients may open eyes but lack evidence of conscious interaction.

Brain death vs persistent vegetative state

Feature	Brain death	Persistent vegetative state
Brainstem reflexes	Absent when formally tested.	Preserved.
Spontaneous breathing	Absent on apnea testing when prerequisites met.	Present.
Legal status	Death in most jurisdictions when criteria met.	Alive with severe disorder of consciousness.
Eye opening	Absent as brain function.	May open eyes; sleep-wake cycles may occur.
Confounders	Must exclude hypothermia, intoxication, metabolic derangement.	Diagnosis also requires careful longitudinal assessment.

9. Integrated cases

Case 1: quiet confusion

An 84-year-old becomes withdrawn and inattentive after hip surgery. No agitation. New urinary retention and oxycodone use are present.

Reasoning: Hypoactive delirium is likely. Treat triggers: urinary retention, pain regimen, sleep disruption, infection search, medication reduction, reorientation.

Case 2: sudden collapse at the gym

A 42-year-old faints during exertion without prodrome. ECG shows LVH and deep Q waves.

Reasoning: High-risk cardiac syncope. Evaluate for structural disease such as hypertrophic cardiomyopathy and arrhythmia risk.

Case 3: shaking after standing

A patient feels warm and nauseated while standing, faints, has three brief jerks, and wakes quickly oriented.

Reasoning: Convulsive syncope is more likely than epilepsy. ECG and syncope risk assessment remain important.

Case 4: first seizure

A 59-year-old has a first focal seizure with right arm jerking followed by weakness. Sodium is normal.

Reasoning: Focal seizure suggests a structural cortical lesion until proven otherwise. Brain imaging and EEG are needed.

Case 5: persistent coma after seizure

A convulsion stops after benzodiazepine, but the patient remains unresponsive for an unexpectedly long period.

Reasoning: Consider nonconvulsive status epilepticus, drug effect, metabolic derangement, hypoxia, or structural injury. EEG is important.

Case 6: pinpoint pupils

A somnolent patient has respirations of 6/min and pinpoint pupils.

Reasoning: Opioid toxicity is likely; support airway/ventilation and give naloxone while continuing broad evaluation.

10. Graph-RAG extraction map

Concept nodes and relationships

Node	Class	Edges to encode
Altered mental status	Clinical syndrome	caused_by metabolic, oxygenation, vascular, endocrine, seizure, trauma, uremia, psychiatric, infectious, drug causes.
Delirium	Clinical syndrome	distinguished_from dementia by acute onset, fluctuation, inattention; treated_by cause correction and nonpharmacologic prevention.
Coma	Clinical syndrome	evaluated_by ABCs, glucose, pupils, motor response, CT, labs; caused_by structural or toxic-metabolic processes.
Syncope	Clinical syndrome	caused_by transient cerebral hypoperfusion; classified_as reflex, orthostatic, cardiac, PE, rare cerebrovascular.
Cardiac syncope	High-risk syndrome	associated_with exertion, supine event, no prodrome, abnormal ECG, structural heart disease.
Seizure	Neurologic event	caused_by abnormal neuronal discharge; classified_as focal or generalized; evaluated_by labs, imaging, EEG.
Status epilepticus	Neurologic emergency	treated_by benzodiazepine, antiseizure loading, ICU/anesthetic escalation if refractory.
Nonconvulsive status epilepticus	Neurologic emergency	presents_as persistent AMS after seizure; diagnosed_by EEG.
Psychogenic nonepileptic seizure	Functional neurologic event	distinguished_by video EEG; requires respectful explanation and mental health/neurology follow-up.
Brain death	Death by neurologic criteria	requires irreversible absence of brain/brainstem function and exclusion of confounders.

11. High-yield traps

- Do not call syncope a seizure solely because there were a few brief jerks.
- Do not call an older quiet patient 'sleepy' without considering hypoactive delirium.
- Do not use a normal early CT to exclude all dangerous neurologic causes of AMS.

• Do not give antipsychotics as primary treatment for delirium; they are symptom-control tools for selected unsafe agitation.

• Do not forget glucose in every AMS, seizure, or transient LOC presentation.

• Do not delay benzodiazepines in convulsive status epilepticus.

• Do not assume persistent unresponsiveness after a seizure is always postictal; consider nonconvulsive status.

• Do not diagnose vasovagal syncope confidently when syncope occurs during exertion, while supine, without prodrome, or with an abnormal ECG.

12. Practice questions

Q1. A patient has acute fluctuating inattention and visual hallucinations after starting diphenhydramine. Diagnosis?

Answer: Delirium from anticholinergic medication effect until proven otherwise.

Q2. What test should every syncope patient receive?

Answer: ECG.

Q3. A patient has transient LOC, rapid recovery, and prodrome of warmth and nausea. Most likely category?

Answer: Reflex/vasovagal syncope.

Q4. A first seizure patient has persistent focal weakness and headache. What is needed urgently?

Answer: Brain imaging to evaluate structural lesion, hemorrhage, or stroke.

Q5. What treatment comes first in ongoing generalized convulsive status epilepticus after glucose/ABCs?

Answer: Full-dose benzodiazepine.

Q6. What should be considered when a patient remains comatose after convulsions stop?

Answer: Nonconvulsive status epilepticus, among other causes.

Q7. What delirium feature most helps separate it from dementia?

Answer: Acute onset with fluctuating attention and awareness.

Q8. What makes exertional syncope dangerous?

Answer: It suggests cardiac obstruction or arrhythmia until proven otherwise.

13. References and clinical cross-check anchors

This chapter was generated as original educational content using the uploaded neuro sections on altered mental status/coma and syncope/seizures as the main coverage guide. Clinical anchors were cross-checked against AAN/AES first seizure guidance, American Epilepsy Society status epilepticus guidance, ACC/AHA/HRS syncope guidance, and American Geriatrics Society/geriatric delirium resources. Local hospital protocols should determine exact seizure medication selection, EEG availability pathways, brain death procedures, and syncope admission criteria.